ARLETTE GELLER

Medical Device Design Engineer

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WORK EXPERIENCES

GE HealthCare, Experience Design Team - Waukesha, WI

Mechanical Engineering Consultant - Developed and executed brand transition strategies, leading to streamlined product development processes. 2022 - 2024

- ***Designed and standardized components*** *to ensure consistency across all imaging modalities.*
- ***Managed material compliance activities,*** *including coordinating with suppliers and overseeing material testing and validation.*
- ***Conducted design reviews*** *and implemented effective supplier qualification processes.*
- ***Produced CAD models and technical drawings*** *on Windchill and PTC Creo Parametric.*
- ***Led the introduction of new materials****, ensuring adherence to engineering specifications and industrial standards.*
- ***Authored and refined engineering specifications and procedures,*** *establishing design requirement standards to meet industrial acceptance criteria.*

Nooshee LLC - Durham, NC

Biomedical Engineer - Designed and developed a cutting-edge breast pump focusing on enhancing mother comfort, milk nutrient preservation, and extraction efficiency. 2022

- ***Led product design efforts,*** *including defining product specifications, user needs, and conducting competitor research.*
- ***Iterated and documented prototype,*** *including CAD models and technical drawings on Fusion360.*
- ***Produced and tested physical prototypes,*** *experimenting concepts to satisfy needs and determine feasibility.*
- ***Enhanced software capabilities,*** *expanding adaptability to different use cases and scenarios.*
- ***Authored patent application*** *to support intellectual property applications.*

Center for Global Women's Health Technologies - Duke University - Durham, NC

Consultant ∞ - Development of "Pocket Colposcope" verification platform and product materials Including animations, renders and CAD models and prototype. Utilized for product showcase, commercialization and research collaborations demonstrations. 2022

Research Assistant ∞ - Redesigned and optimized colposcope platform for cervical biopsy HSP90 staining and analysis to allow robustness and portability for hospital settings and use case. 2020 - 2021

Calla Health - Durham, NC

Junior Engineer ∞ - Engineered design improvements for handheld colposcope, developed reprocessing procedures and packaging. 2022

- ***Requirements verification*** *to develop validation activities to ensure quality and performance of manufactured devices.*
- ***Led the development of operator and standardized training*** *activities, ensuring comprehensive understanding and effective use of the equipment.*
- ***Supported grant writing, marketing, and branding activities,*** *contributing to the commercialization of the product.*

Biomedical Engineering Intern ∞ 2021

- ***Designed a portable testing platform*** *to evaluate device specifications and ensure quality compliance.*
- ***Developed packaging, labeling, and user manuals,*** *contributing to the overall product presentation and usability.*
- ***Conducted material analysis throughout the product life cycle,*** *including reprocessing assessments to optimize product durability and performance.*
- ***Led the market research,*** *conducted in-depth analysis of target markets to drive strategic decision, market positioning and business plan.*

Research Lab - Eric S. Richardson Ph.D. - Duke University - Durham, NC

Validation Study ∞ - Design, execution and publishing physiological and particulate validation studies for orthopedic surgical helmet manifold modification efficacy for COVID-19. 2021

- "Helmet Modification to PPE With 3D Printing During the COVID-19 Pandemic at Duke University Medical Center: A Novel Technique" ∞

Independent Study - Designed technology to tackle negative impacts of surgical masks during long-term use in the OR. 2020 - 2021

- *Designed and developed an internal support structure and external accessory to promote airflow and improve the comfort and fit of surgical masks.*

EDUCATION

Duke University - Durham, NC

Master of Engineering in Biomedical Engineering ∞ - Pratt School of Engineering - Scholarship and Research Awards 2021

Medical Device Design Certificate ∞ - Department of Biomedical Engineering 2021

Arizona State University - Tempe, AZ

Bachelor of Science in Biomedical Engineering ∞ - Ira A. Fulton School of Engineering - Cum Laude 2020

PROJECTS AND LEADERSHIP

RL3T LLC

Adaptive and Radially Expanding Speculum ∞ - Developed technological solution to enhance ergonomics, improved patient comfort, and increased practitioner efficiency. Design that free up practitioners' hands for inspection and tool use, while also addressing cost-effectiveness, infrastructural reprocessing challenges, and limitations in the field of view and vaginal wall hindrance. 2022

Duke University

Blue Devils Bridges - Participated as a mentor to BME graduate students providing support on professional development activities. 2023

Ceremony Speaker ∞ - Engineering Master's Hooding Ceremony 2022

Engineering Master's Student Council - Founding Member, Vice-President and President 2020 - 2021

Design+Health Program

Engineering Fellow ∞ - Contributed to needs finding, concept selection, product design and development, and patent application preparation for future device implementation at Duke Hospital. Designed and developed a method for managing fluid collections, enhancing patient comfort and recovery through improved intermittent drainage. 2020 - 2021

Advanced Manufacturing and Prototyping

Speculum ∞ - Designed silicon and disposable speculum able to adapt to different body sizes and characteristics. 2020

Mitral Valve Sewing Ring ∞ - CAD model based on MRI segmentation and reconstruction for ring annuloplasty procedure.

Cystoscope ∞ - Patient comfort focused design with large scale manufacturing features and requirements for implementation of photon engine technology.

Arizona State University

Capstone Project

At-home Breast Cancer Screening Device ∞ - Developed portable and low cost breast cup model for routinely screening through impedance measurements and historical data analysis. 2019 - 2020

Instrumentation for Biomedical Engineers

High Spinal Cord Injuries Assistive Technology ∞ - Built device to allow individuals use the computer on their own through a headband for screen navigation, and a mouthpiece as the click functionality. 2019

Microcomputing for Biomedical Engineers

Pen Plotter Machine ∞ - Built machine that Interprets voice into text creating a custom typography on paper. 2019

Product Design and Development III

Vagus Nerve Stimulation Device ∞ - Designed non-invasive ear lobe vagus nerve stimulator system that activates when the subject is performing the desired exercise by detecting body movements through an infrared sensor for remote stroke patients' rehabilitation. 2019

Locomotion Research Lab

Research Assistant - Contributed in gait data acquisition, processing and analysis. 2017 - 2019

- Co-Author **"Nonlinear Evaluation of Gait in Older Fallers and Non-Fallers" ∞**

Biomedical Engineering Society - Mentored younger students with a industry career path interest. 2017 - 2019

Most Interesting Project Award - Inflatable birthing cushion design for at-home labor in the DRC. 2017

CITI Program

Engineers (RCR) - Responsible Conduct of Research 2017

Human Research (IRB) - Social and Behavioral Research 2017

Rotary Club - Lima, Peru

Operation Smile - Medical translator, interpreter and facilitator during surgery, recovering instructions and follow up appointments of cleft lip and palet reconstructive surgery procedures. 2016

PATENTS AND INVENTIONS

RL3T LLC - US20230218157A1 ∞ - Adaptive and Radially Expanding Speculum - Owner and Inventor 2022

Nooshee LLC - US20230405193A1 ∞ - Breast pump system and components and uses thereof - Inventor 2023

Design+Health Program - 63/181,460 - Device and Method of Managing Fluid Collections - Inventor 2021

SOFTWARE AND TOOLS

Slicer, SolidWorks, Shapr3D, Creo, Windchill, Fusion360, Blender, QuickField, FEA, SPSS, G*Power, Photoshop, Illustrator, Inkscape, CAPA, LTSpice, LabView, Arduino, Integrated-Circuit Design, Anaconda, MathCAD, MATLAB, C++, Python, GitHub, Visual Studio Code, ASME, ISO, CIE, Color iQC, REACH, RoHS, MyWorkshop, Anthesis